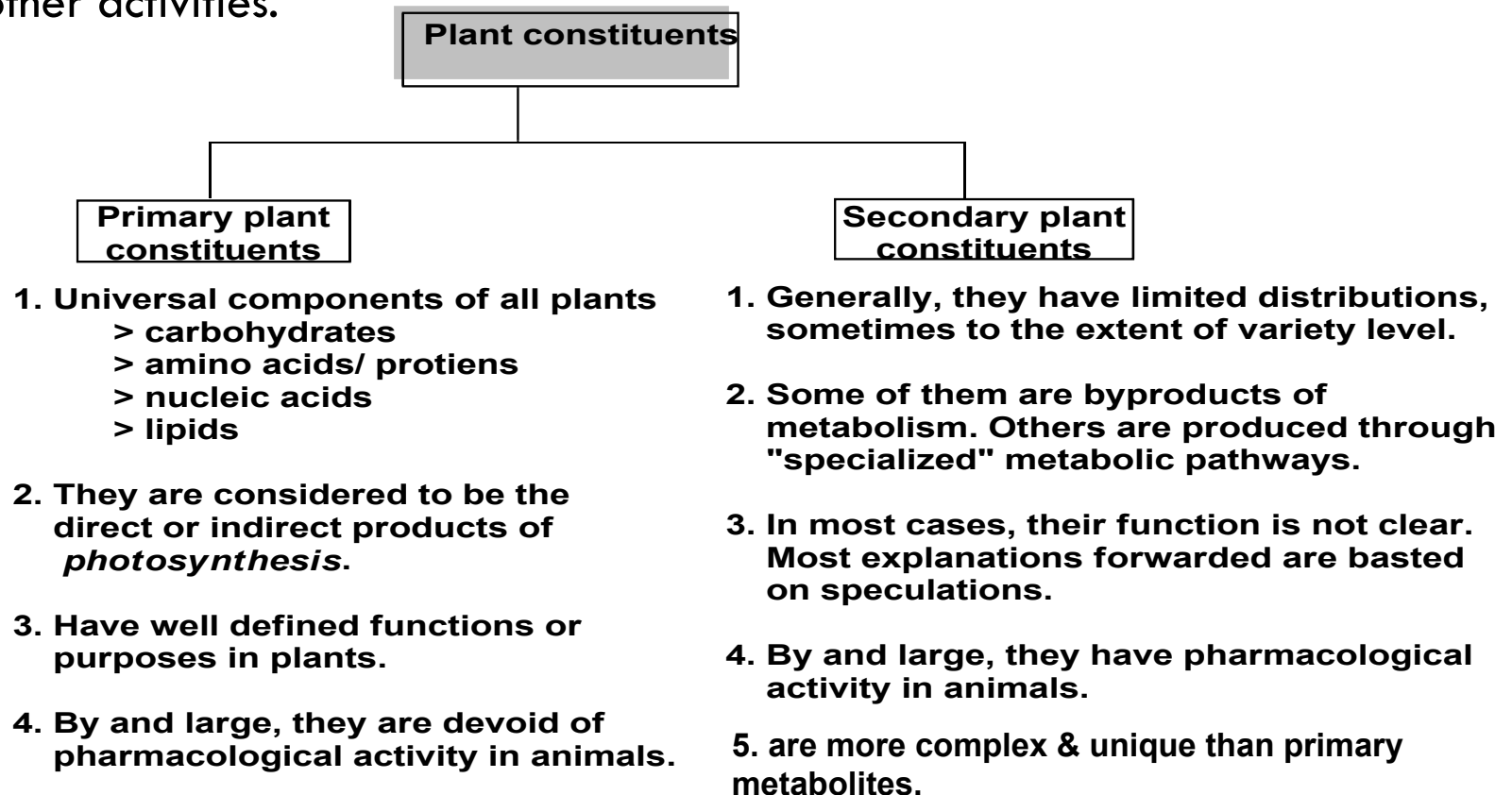


Glycosides

Objectives

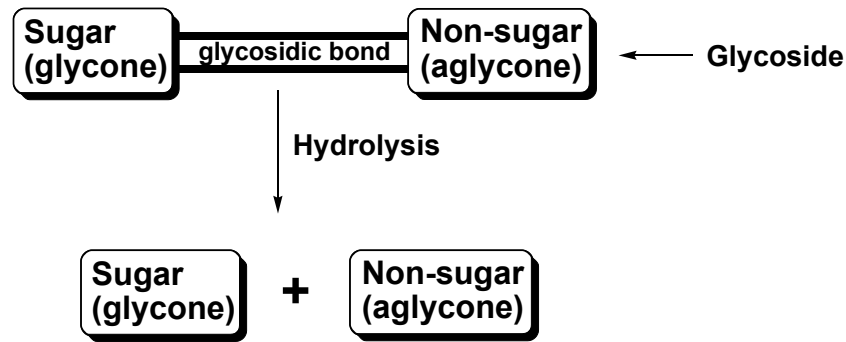
- At the end of this chapter students are expected to:
- Differentiate between primary and secondary metabolites
- Define glycoside
- Mention the physical and chemical properties of glycoside
- List the different types of glycosides
- List the pharmaceutical values different types of glycosides
- Define tannins
- Differentiate between the different types of tannins

- **Primary metabolites** are the compounds that are directly involved in the metabolic pathways of an organism necessary for its growth, development, and reproduction.
- **Secondary metabolites** are the organic compounds that are produced by various organisms that are not directly involved in the growth, development, or reproduction of the organism but are essential in the ecological and other activities.



Glycosides

- is an organic compound, usually of plant origin, that is composed of a sugar portion linked to a non-sugar moiety.
- The glycosides are products of the secondary metabolism of plants by the condensation of a sugar with other organic molecules, in which the linkage alpha or beta is made with the hemiacetal hydroxyl.
- the sugar portion is called glycone, while
- the non-sugar portion is called aglycone or genin.



- Many glycosides occur in plants, often as flower and fruit pigments; for example, anthocyanins.
- Many plants store chemicals in the form of inactive glycosides.
- These can be activated by enzyme hydrolysis, which causes the sugar part to be broken off, making the chemical available for use.
- Therapeutic activity is due to the aglycone.
- The sugar part influences pharmacokinetic and pharmacodynamic property.

Distribution and Function

Distribution

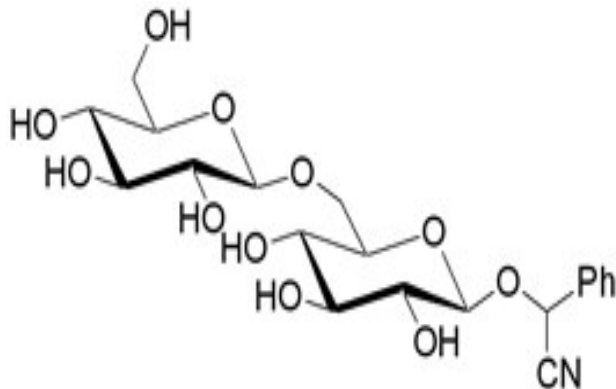
- Plants, Microorganisms, Anthropods...
- In plants they are found in varying amounts in seeds, fruits, roots, barks, & leaves.
- In some cases , two or more glycosides are found in the same plant.
- It will be noted that such reactions are reversible and in plants glycosides are both synthesized and hydrolysed under the influence of more or less specific enzymes
- Glycosides often are accompanied by enzymes that are capable of synthesizing or hydrolyzing them.
- Because of the diversity of aglycones and glycosides, these compounds do not share a common biosynthetic pathway.

Classification

- By glycone/presence of sugar
 - If the glycone group of a glycoside is glucose, then the molecule is a glucoside;
 - if it is fructose, then the molecule is a fructoside;
 - if it is glucuronic acid, then the molecule is a glucuronide; etc.
- Atom from the aglycone involved in the glycosidic linkage (By linkage):

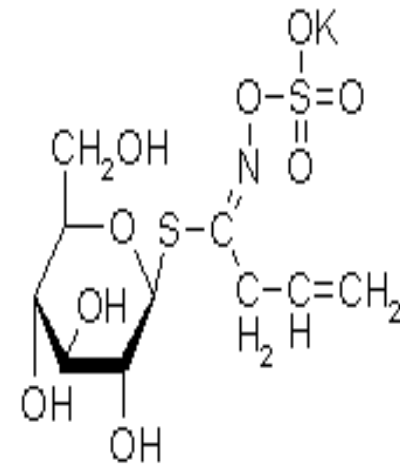
● Aglycone- O- Sugar	O-glycosides
● Aglycone- C- Sugar	C-glycosides
● Aglycone- S- Sugar	S-glycosides
● Aglycone- N- Sugar	N-glycosides

- a glycoside is any molecule in which a sugar group is bonded through its anomeric carbon to another group via a glycosidic bond.



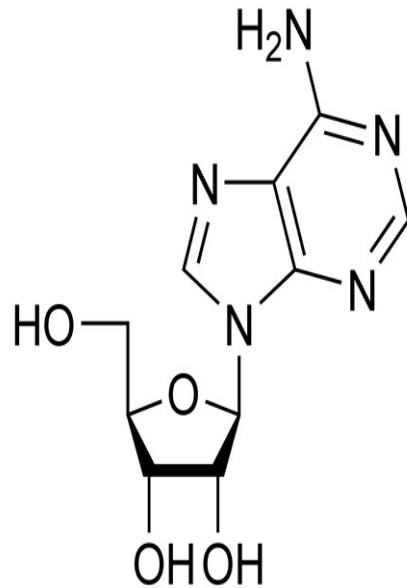
O-glycosides

- are the most abundant glycosides
- easily be hydrolyzed by boiling in mineral acids



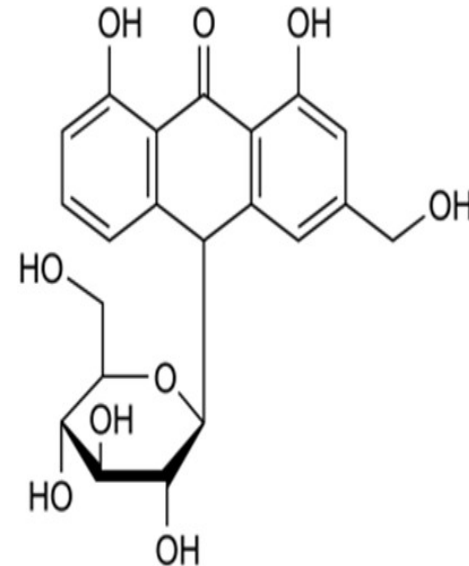
S-glycosides

- Natural S-glycosides are very few
- present in glucosinolates (thioglycosides)



N-glycosides

- easily be hydrolyzed by boiling in mineral acids
- present in nucleosides



C-glycosides

- The sugar is condensed directly to a carbon atom
- are highly resistant to acid hydrolysis
- "nonhydrolysable by acids or enzymes"

By nature of the glycoside

- Primary glycosides: Originally present in the plant
e.g. Purpurea A
- Secondary glycosides: Resulted from removal of one sugar from the primary glycosides e.g. Digitoxin

By aglycone

- Glycosides are also classified according to the chemical nature of the aglycone.
 - Anthraquinone glycosides
 - Coumarin glycosides
 - Cyanogenic glycosides
 - Flavonoid glycosides
 - Phenolic glycosides
 - Saponins glycosides
 - Cardiac glycosides

Physical properties

- Most are crystalline
- Colorless, solid, amorphous, nonvolatile (flavonoid- yellow, anthraquinone-red or orange.
- Most have bitter taste except: populin, glycyrrhizin, stevioside
- Odorless except saponin (glycyrrhizin)
- Glycosides are water soluble compounds and insoluble in the organic solvents.
 - Glycone part: water soluble, insoluble in the organic solvents.
 - Aglycone part: water insoluble, soluble in the organic solvents

Chemical properties

- D-glucose is the most common sugar
 - More than one sugar may be present
- Most glycosides are hydrolysable
- Glycosides are easily hydrolyzed by mineral acids, water and enzyme.
- Acetal linkage between the aglycon and glycone more unstable than that between two individual sugars within the molecule.
- They show optical activity normally they are levorotatory
- Reduce Fehling's solutions only after hydrolysis (They are non-reducing).

Precautions before extraction

- Glycosides often are accompanied by enzymes that are capable of synthesizing or hydrolyzing them.
- This phenomenon introduces problems in the isolation of glycosides because the disintegration of plant tissues, with no precautions to inhibit enzymatic activity leads in some cases, to partial or complete hydrolysis of the glycosides
- The enzymes first should be damaged to extract these glycosides by the following steps.

•

Maintenance of neutral conditions:

- Neutral pH should be assured before and during extraction because: Acidity may result in hydrolysis.
- This is overcome by addition of CaCO_3
- Defatting of fat-rich organs (e.g. seeds) before extraction:
- High amounts of lipids hinder glycoside extraction usually carried with petroleum ether

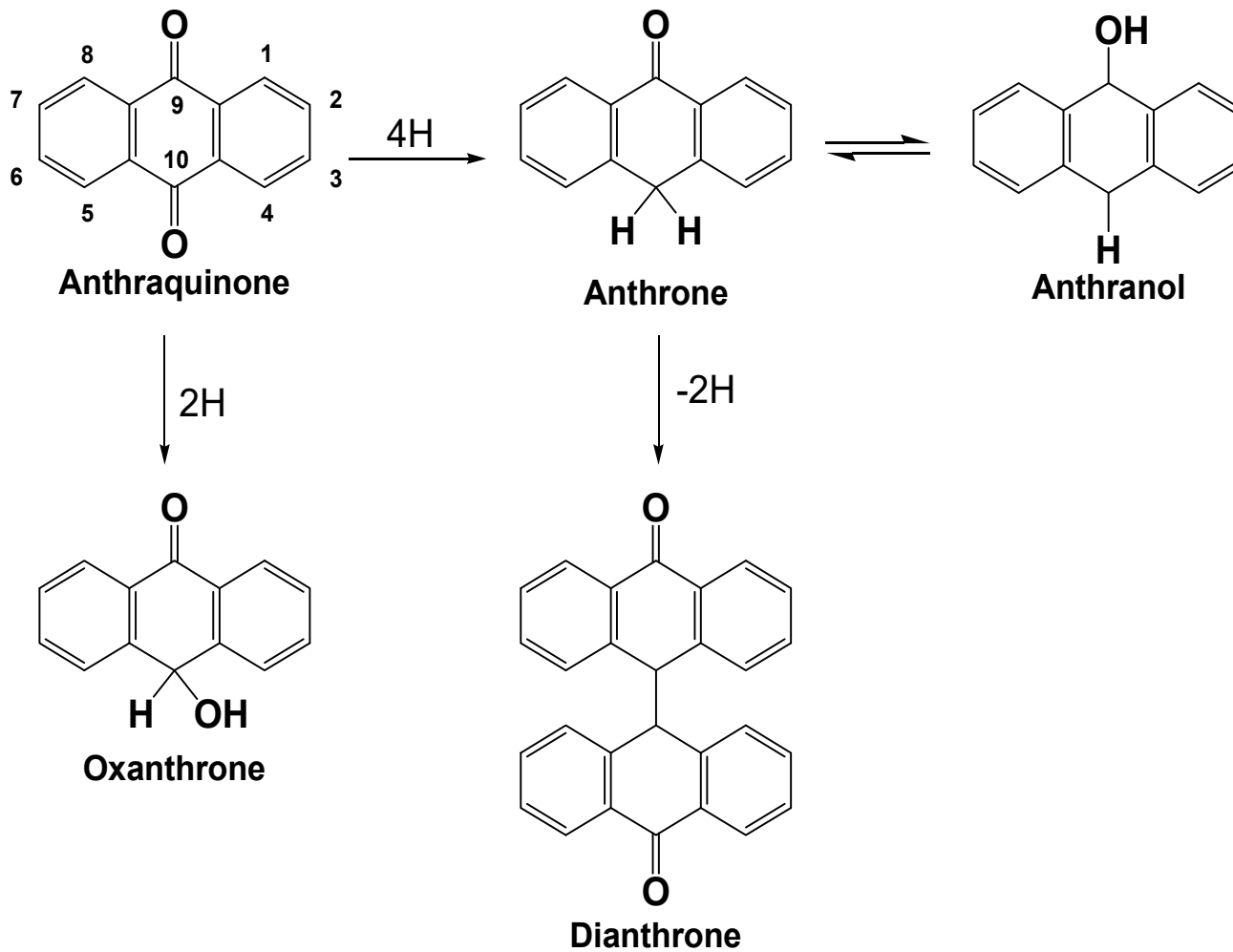
Extraction

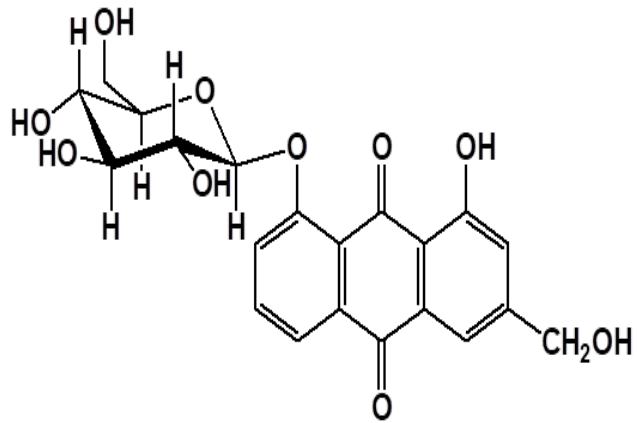
- Extraction of glycosides from plant material (previously dried and properly stored to avoid decomposition of glycosides by plant enzymes) is usually performed with the use of polar solvents (water, hydro-alcoholic solutions, and alcohols).

Pharmaceutical important glycosides

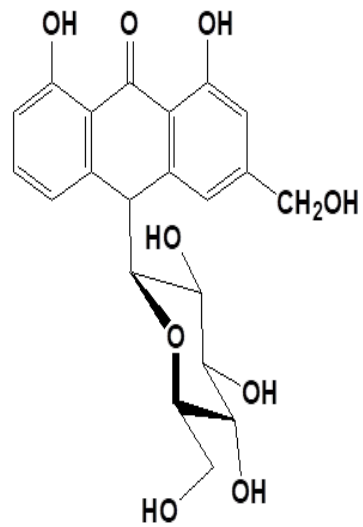
Anthraquinone glycosides

- Anthraquinone glycosides are generally orange, red, or brown-red compounds found in fairly limited distribution within the plant kingdom.
- They are mainly found in dicot plants except the family Liliaceae which are, rhubarb and Aloe species.
- Although some aglycones are absorbed within the small intestines, many of the intact glycosides make it to the colon before being hydrolyzed.
- These glycosides contain an aglycone group that is a derivative of anthraquinone.
- They have a laxative effect.





Aloe-emodin 8-glucoside
(an O-glycoside)



Barbaloin
(a C-glycoside)

Pharmacological use

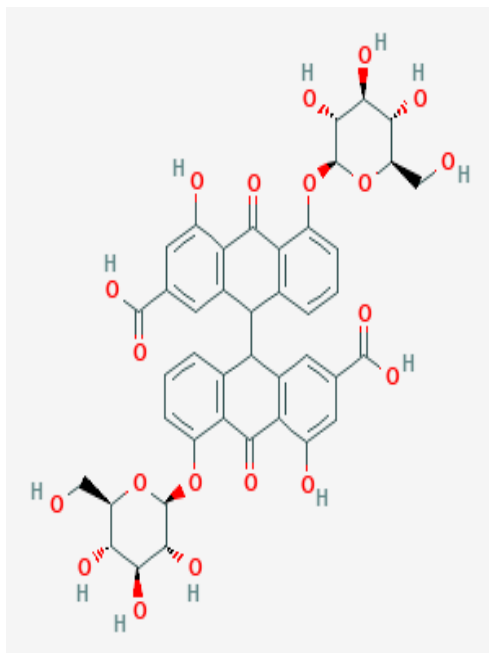
- Used as laxatives or cathartics,
 - Anthranols and anthrones elicit a more drastic action
 - The glycosides are hydrolyzed by the enzymes of the microflora.
 - They act on the large bowel.
 - Their effect may be delayed up to 6 hrs or longer.
 - They increase the strength of peristaltic contractions but not the frequency.
 - At larger doses, they cause colic pain.
 - They are habit forming; they must not be prescribed for more than 15 days.
- anti-inflammatory, antibacterial, antifungal and also as natural dyes

Crude drugs containing anthraquinone glycosides

Senna

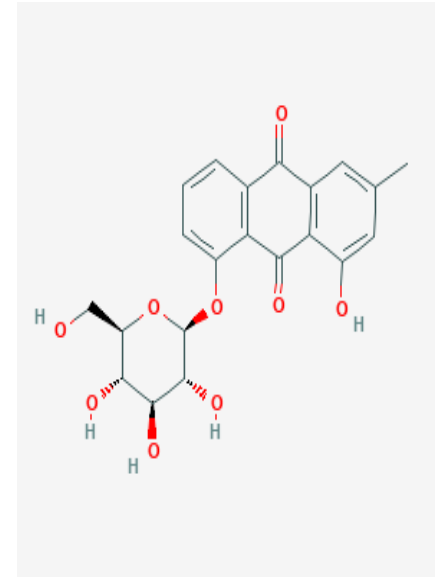
Senosides A and B

Therapeutic activity; Stimulant
cathartics

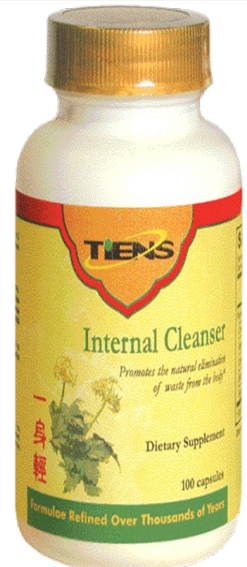


Rhubarb

chrysophanol glycosides
Laxative



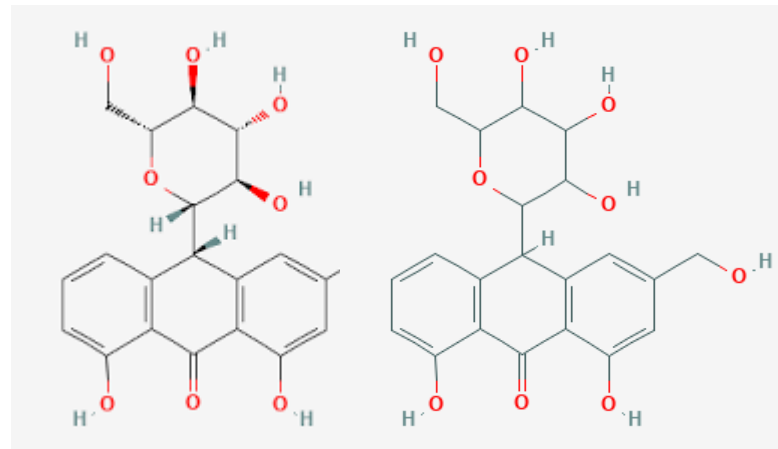
In traditional Chinese medicine, Herbs containing anthraquinone glycosides, particularly *Rheum palmatum* (rhubarb), were administered boiled if noncathartic properties were preferred because this would hydrolyze many of the anthraquinone glycosides.



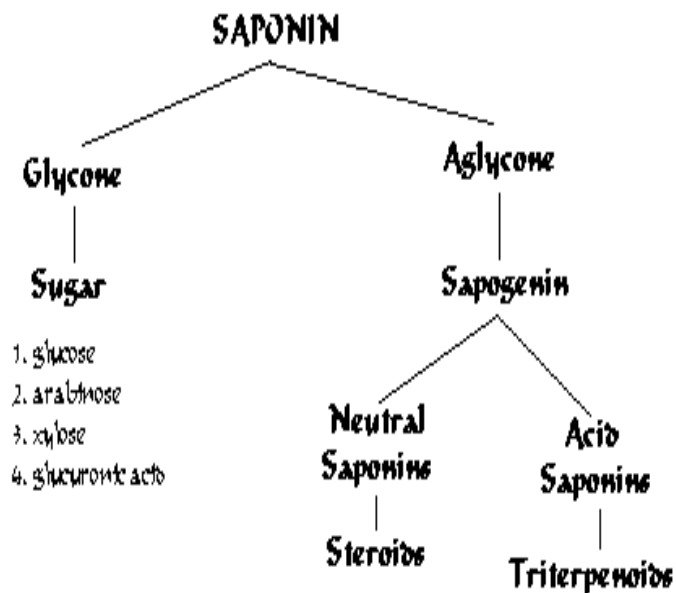
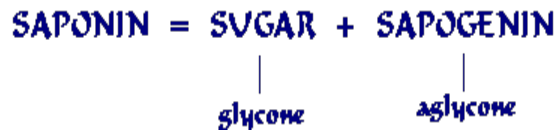
Aloe

The major anthraquinone glycosides present in the exudate of Aloe species include aloins (barbaloin and isobarbaloin), known to release aloe-emodin and aloe-emodin-9-anthrone in the intestines.

They are well-known as potent laxative compounds and pharmacological studies have shown that aloin possesses anticancer



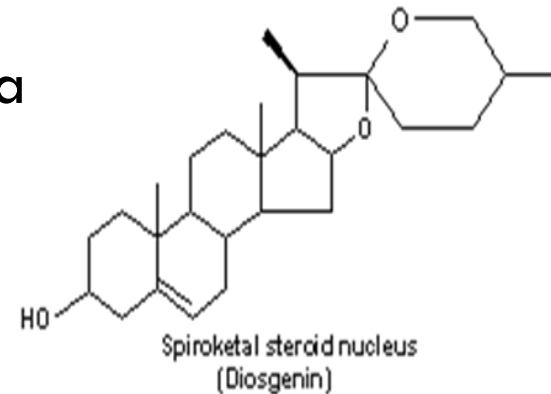
SAPONIN GLYCOSIDES



- From the Latin word “sapo,” (Eng. “soap”), reflecting their surfactant properties: ability to form stable soap-like foams in aqueous solutions.
- have long been used in many parts of the world for their detergent properties
- E.g. Causes haemolysis of red blood cells hence toxic if injected to blood stream

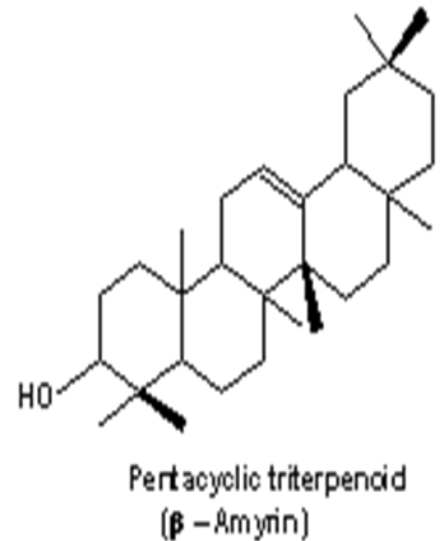
Steroidal Saponins

- Steroidal saponins consist mainly of a C-27 spirostane skeleton, generally comprising of a six-ring structure.
- Less widely distributed in nature
- Common in:
 - Dioscoraceae (e.g. discoria spp),
 - Amaryllidaceae (e.g. Agavae spp)
 - Liliaceae (e.g. yucca and trillium spp)
- useful starting materials for the synthesis of steroidal drugs
- Diosgenin, Hecogenin, Botogenin and others



Triterpenoidal Saponins

- Aglycone consist of a pentacyclic C-30 skeleton
- Abundant in many dicotyledonous families:
 - Caryophyllaceae, Sapindaceae, Polygalaceae



Uses

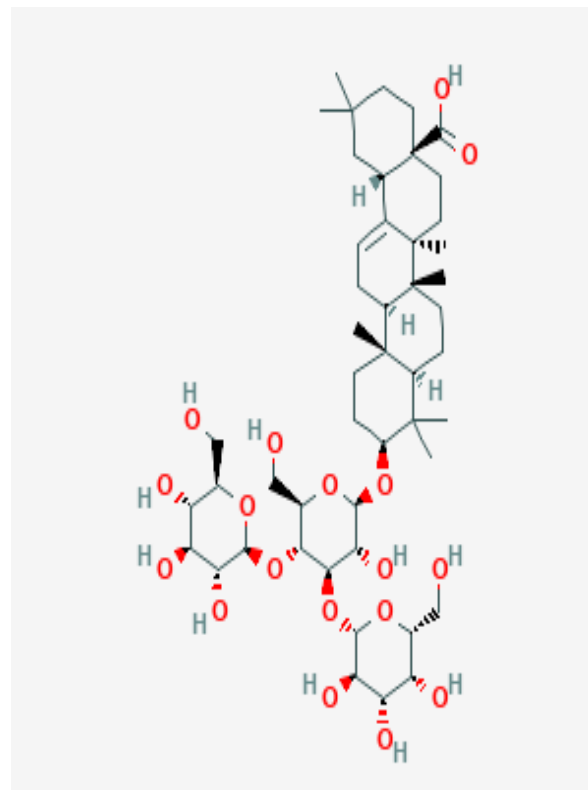
- Cleaning industrial equipment and fine fabrics
- Powerful emulsifier of certain resin, fats and fixed oils
- Antifungal and antibacterial agents
- Lowering of blood cholesterol
- Inhibition of cancer cell growth

Crude Drugs Containing Saponin Glycosides

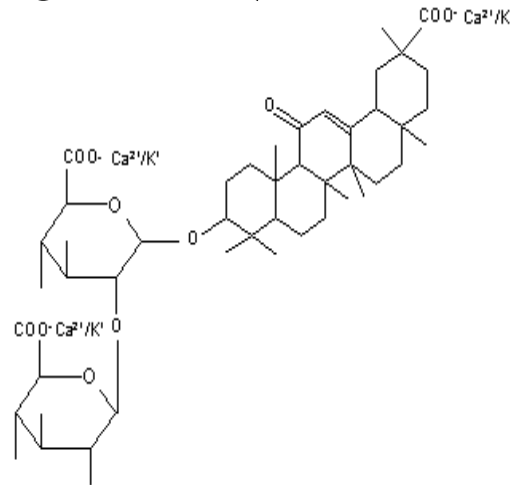
- *Phytolacca dodecandra*



Lemmatoxin:
molluscicide to control schistosomiasis.



- Licorice (*Glycyrrhiza glabra*)

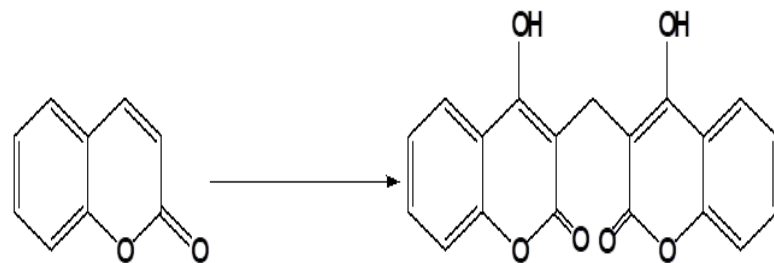


Glycyrrhizin = Ca²⁺ & K⁺ salts of Glycyrrhizinic acid

- Contains a saponin glycoside called Glycyrrhizin
- inhibits liver cell injury
- Antiinflammatory
- used in the food and beverage industry due to their distinctive sweet and typical licorice flavor.

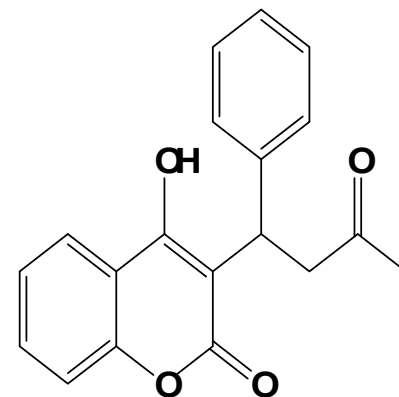
Coumarin glycosides (lactone glycosides)

- Here, the aglycone is coumarin or a derivative.
- They are widely distributed in nature
 - Leguminosae, Rubiaceae, Umbelliferae, Solanaceae
- Dicoumarol is very potent anticoagulant which can not be used for treatment purpose.
- As a lead compound for the synthesis of coumarin based oral anticoagulants such as warfarin.
- are competitive inhibitors of vitamin K in the biosynthesis of functional thrombine.



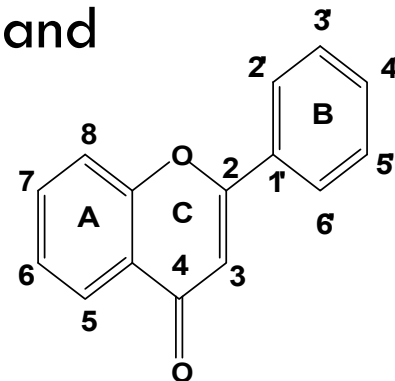
coumarin
does not have
anticoagulant activity

Dicoumarol (bishydroxycouma
have strong
anticoagulant activity

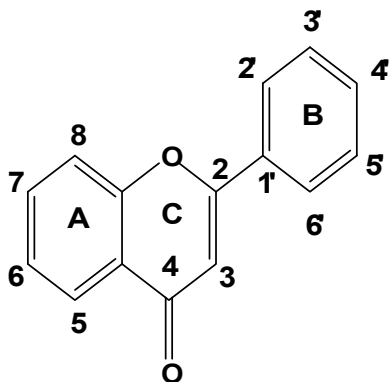


Flavonoid glycosides

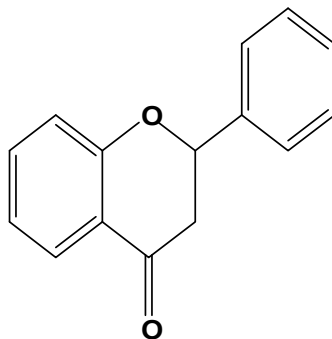
- Flavonoids share a central three-ring structural motif and are all synthesized from cinnamic acid.
- the aglycone has a $C_6C_3-C_6$ nucleus
- are usually substituted by hydroxyl groups
- are responsible for the color of flowers, fruits and leaves.
- Many have yellow color
- Anthocyanins have red, blue or purple color
- Most absorb in the UV region
- Exhibit various pharmacological activities :
 - Anti-inflammatory (and analgesic) properties
 - Anti-allergic effects
 - Tumour inhibition promotion
 - Antioxidant activity



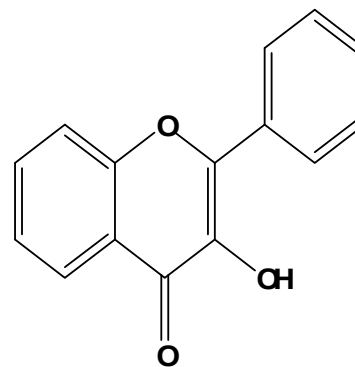
Structures



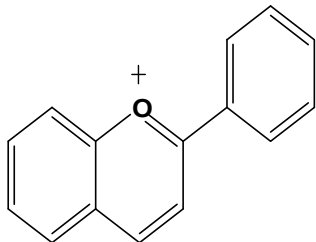
Flavone



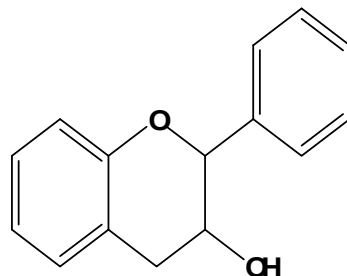
Flavanone



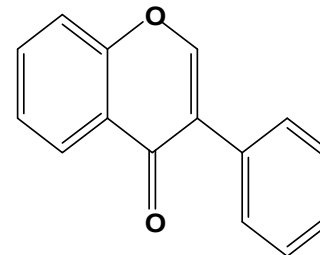
Flavonols



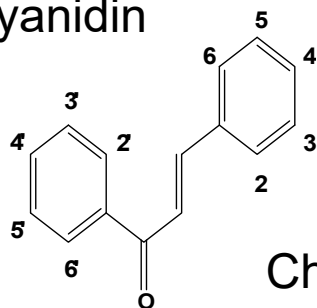
Anthocyanidin



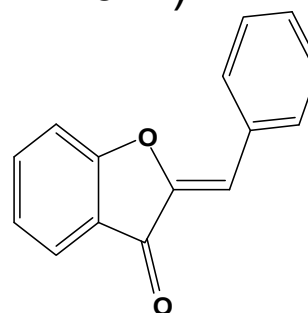
Catechin (flavan-3-ol)



Isoflavanoid



Chalcone

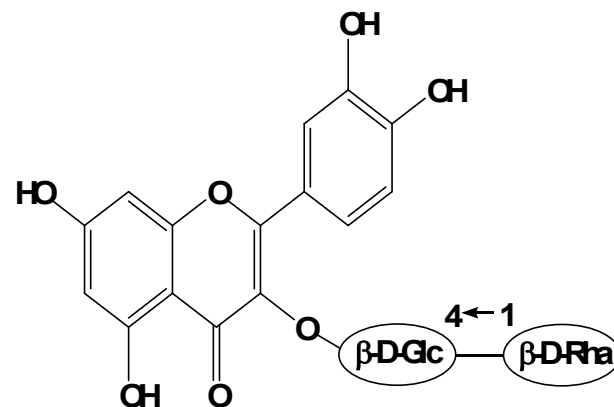


Aurones

Some Medicinally Important Flavonoids

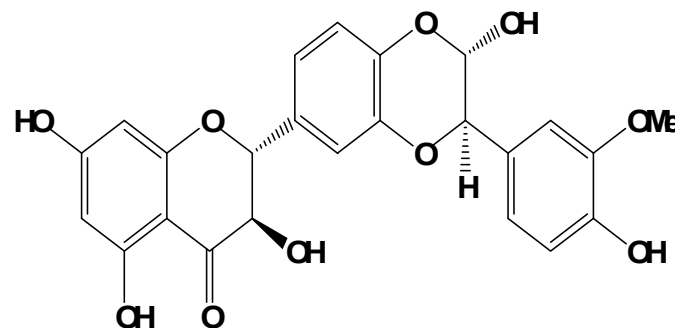
Rutin

- Used as for the treatment of capillary bleeding and increased capillary fragility
- buckwheat, Japanese pagoda tree, and Eucalyptus



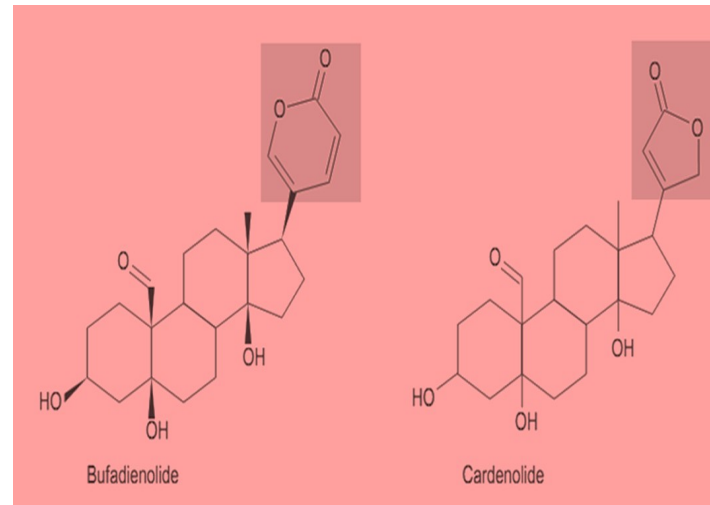
Silybin

- It is the major active compound isolated from silymarin.
- a hepato-protective Flavonoidal fraction obtained from seed/fruits of *Silybum marianum* (Asteraceae) (Milk Thistle, St. Mary Thistle)



Cardiac glycosides

- Plant glycosides with specific action on heart.
- Historical sources and uses
 - South American toad skins as arrow poisons to assassinate people.
- Aglycone : steroid nucleus
- Sugar: monosaccharide or a polysaccharide with β -1,4- glucosidic linkages

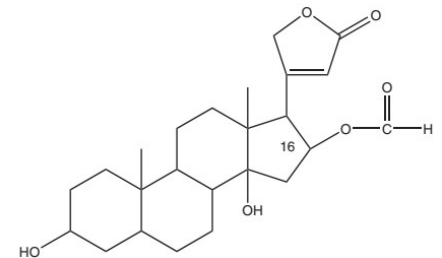


- Scrophulariaceae
 - *Digitalis purpurea* leaves (fox glove)
 - *Digitalis lanata* leaves – white flowers
- Apocyanaceae
- Liliceae
- Modern use:
 - to treat congestive heart failure.



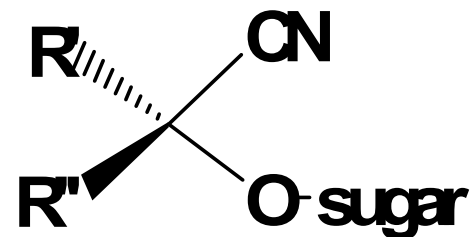
Digitoxin:

- is widely used in the treatment of various heart conditions, namely atrial fibrillation, atrial flutter and sometimes heart failure that cannot be controlled by other medication
- The main secondary glycoside in *Digitalis purpurea*.

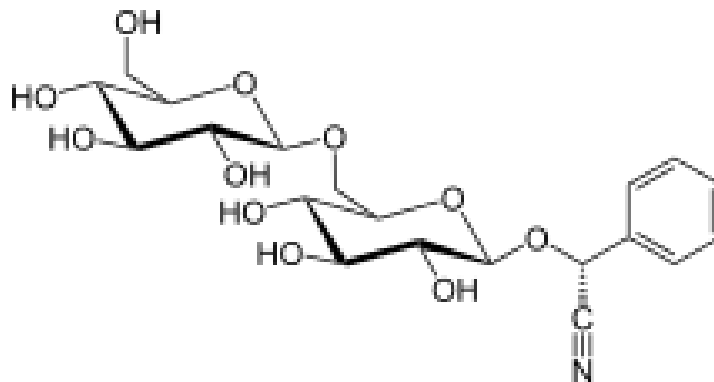


Digitoxigenin, a cardenolide aglycone

Cyanogenic glycosides



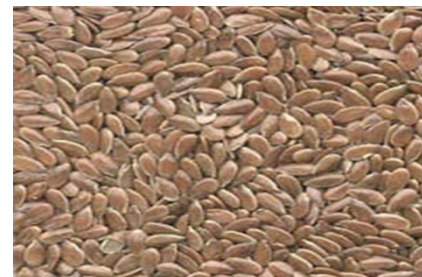
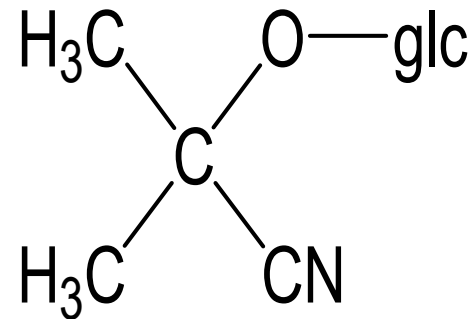
- Cyanogenic glycosides have amino acid–derived aglycones that are mostly a safety concern in medicinal plants.
- Cyanogenic glycosides are natural plant toxins.
- Cyanogenic glycoside is not toxic on its own
- Cyanide is formed following the hydrolysis of cyanogenic glycosides that occur during crushing of the edible plant.
- Over 2500 plant species
 - Family Rosaceae, Fabaceae, Poaceae etc.



- The concerns about cyanogenic glycosides are twofold.
- First, these compounds interfere with iodine organification and thus can cause or promote goiter and hypothyroidism.
- However, this problem has been clearly documented to occur only in the setting of iodine deficiency or with massive overconsumption.
- Second, cyanogenic glycosides do spontaneously degrade to release potentially lethal hydrogen cyanide once the glycosidic linkage is hydrolyzed.
- However, unless massive and rapid intake of cyanogenic glycosides occurs, the cyanide is quickly and safely detoxified by hepatic thio sulfate sulfurtransferase.

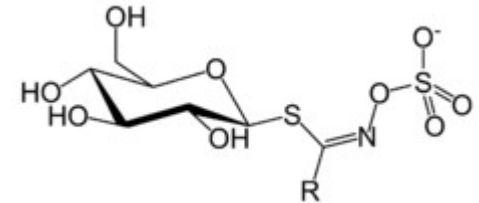
Linamarin

- is a cyanogenic glucoside found in the leaves and roots of plants such as cassava, lima beans, and flax.
- It is a glucoside of acetone cyanohydrin.
- has a molluscecidal activity.
- cyanogenic glycosides have role in cancer treatment

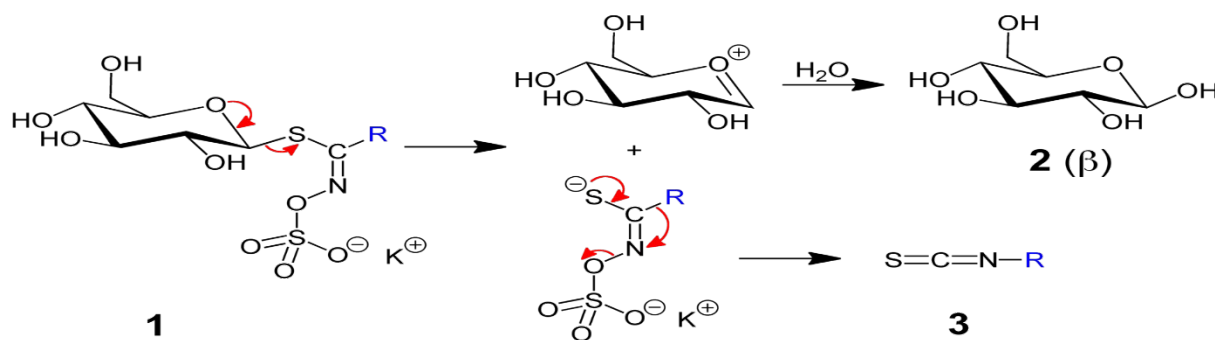


Glucosinolates

- Glucosinolates yield pungent, sulfur-containing, amino acid–derived aglycones, such as glucobrassicins, thiocyanates, and isothiocyanates, that give Brassicaceae, Capparidaceae, Resedaceae, and other families of plants their unique flavors. (including broccoli, cabbage, [cau flower](#), rapeseed, mustard)
- The glucosinolate itself is not pungent. When it encounters the enzyme myrosinase, normally stored in another compartment of the cell, the aglycone is formed which then usually rearranges into a pungent and corrosive isothiocyanate.



- Similar to other glycosides, glucosinolates are water soluble, but their aglycones are hydrophobic. These aglycones have an oily consistency and are often known as mustard oils.
- Degradation products of glucosinolates include other [organosulfur compounds](#), such as the [isothiocyanates](#) and dithiothiols.
- Despite their strong odor and oily nature, these compounds are chemically very different from volatile oils and should not be confused with them.
- Glucosinolate aglycones are traditionally used topically as rubefacients. They cause a warming sensation and reddening related to dilation of dermal capillaries



Reading assignment

- Aldehydic glycosides
- Alcohol glycosides
 - What are they
 - Their distribution
 - Their pharmaceutical use

Tannins

Introduction

- The term 'tannin' was first applied by Seguin in 1796 to denote substances present in plant extracts which were able to combine with protein of animal hides,
 - prevent their putrefaction and convert them into leather.
- Holding tannins in the mouth produces a strong sensation of astringency.
- It is believed that tannins bind to proteins in the tongue and mucous membrane in the oral cavity and then induce their denaturation.
- This type of denaturation of proteins in the oral mucosa by tannins is called a shrinking effect

- chemically , tannins are complex substances that usually occur as a mixtures of polyphenols
 - that are difficult to separate because they do not crystallize.
- This definition excludes simpler phenolic substances, often present with tannins, such as gallic acid, catechins and chlorogenic acid,
- Such substances of relatively low molecular weight are called **‘pseudo-tannins’**.
- Most true tannins have molecular weights from 300 to 3000.
- detected
 - qualitatively by a tanning test (the goldbeater’s skin test)
 - quantitatively by its adsorption on standard hide powder.

Occurrence

- Tannins are of wide occurrence in plants and are usually found in greatest quantity in dead or dying cells.
- Tannins dominate quantitatively the vast plant secondary metabolites, being the fourth most abundant compounds in the plant vascular tissue next to cellulose, hemicellulose, and lignin
- common to vascular plants, exist primarily within woody tissues but can also be found in leaves, flowers, or seeds.
- Lower plants such as algae, fungi and mosses do not contain much tannin.

- Unripened fruits are high in tannin content.
- The high tannin content discourages fruit eating animals from consuming the fruit until the seeds are mature and ready for dispersal. As the fruit ripens the tannin content lessens.
- Beside fruits, tannins are also contained in coffee, tea, red wine and beer.
- The initial astringent taste when you sip a red wine actually comes from tannins in the wood of the oak barrels in which the wine was aged.
- Tannins are also responsible for many of the enchanting colors seen in flowers and the final beauty of autumn leaves.

Classification

Tannins

True
tannins

Pseudo
tannins

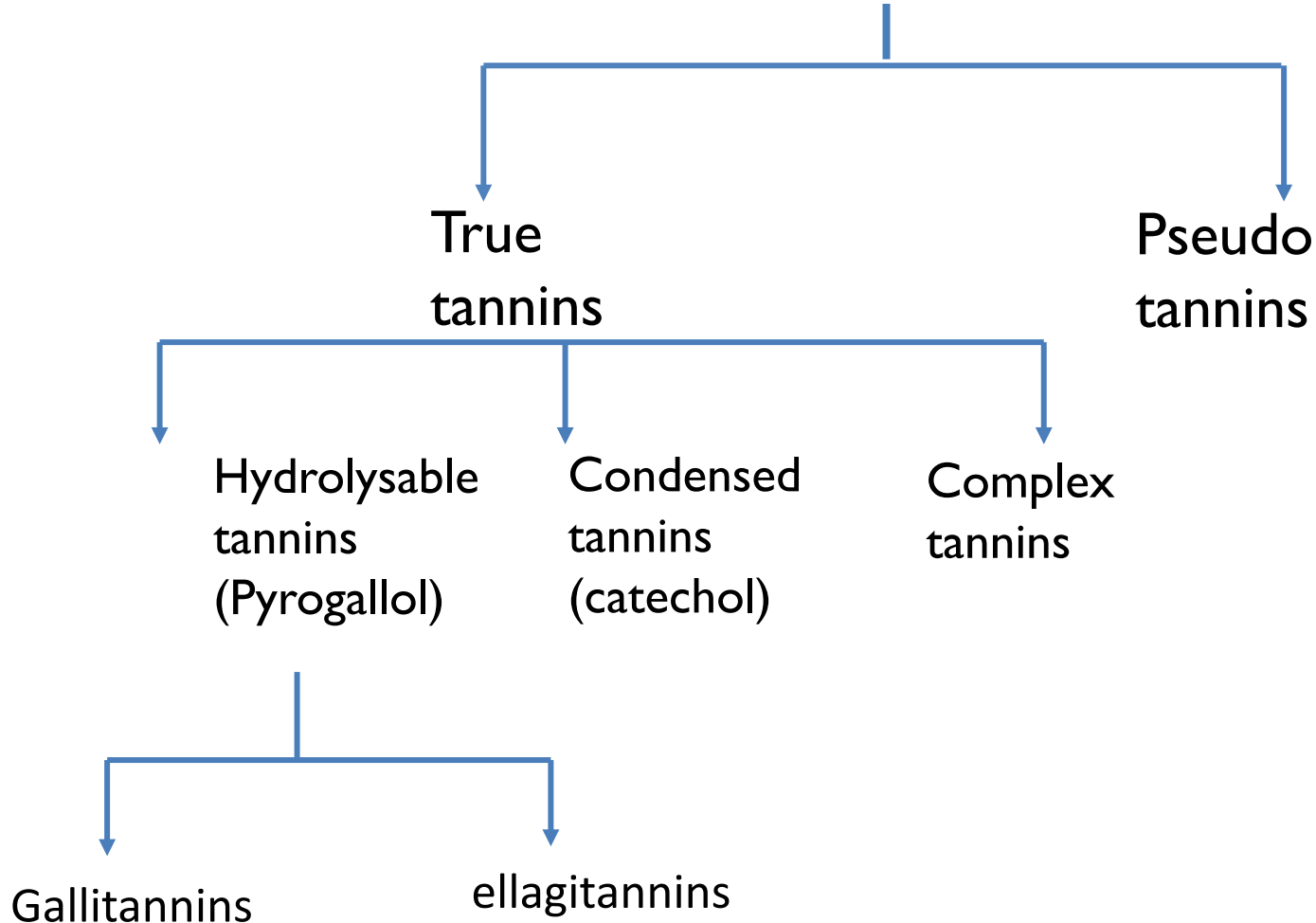
Hydrolysable
tannins
(Pyrogallol)

Condensed
tannins
(catechol)

Complex
tannins

Gallitannins

ellagitannins



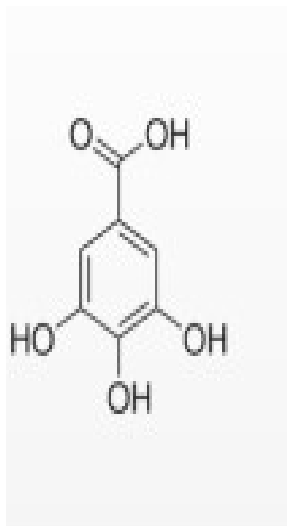
True tannins

- polyhydroxy phenolic compounds with molecular weight ranging from about 300 to 3000.
- Besides giving the usual phenolic reactions,
- Have special properties such as the ability to precipitate alkaloids, gelatin and other proteins.
- The classical division of tannins was based on their resistance or not, to hydrolysis in the presence of hot water or the enzymes tannases.

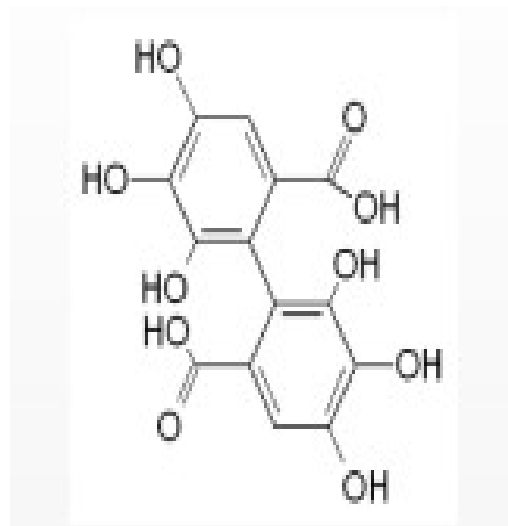
Hydrolysable tannins (pyrogallol)

- These may be hydrolyzed by acids or enzymes such as tannase.
- They are formed from several molecules of **phenolic acids** such as **gallic** and **hexahydroxydiphenic acids** which are united by **ester linkages** to a **central sugar (mainly glucose)** molecule.
- The hydroxyl groups of the carbohydrate are partially or totally esterified with phenolic acids
- When the phenolic acid is
 - gallic acid: gallotannins
 - hexahydroxydiphenic acid: ellagitannins
- Like gallic acid their solutions turn **blue** with **iron salts**.

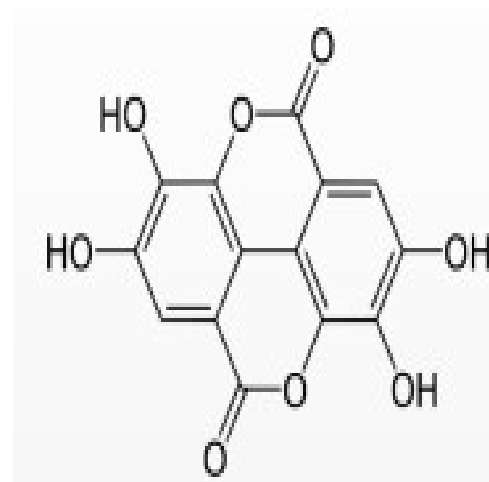
- They were formerly known as **pyrogallol tannins**, because on **dry distillation** gallic acid and similar components are converted into **pyrogallol**.
- Two principal types of hydrolysable tannins are
 - gallitannins and ellagitannins



Gallic acid



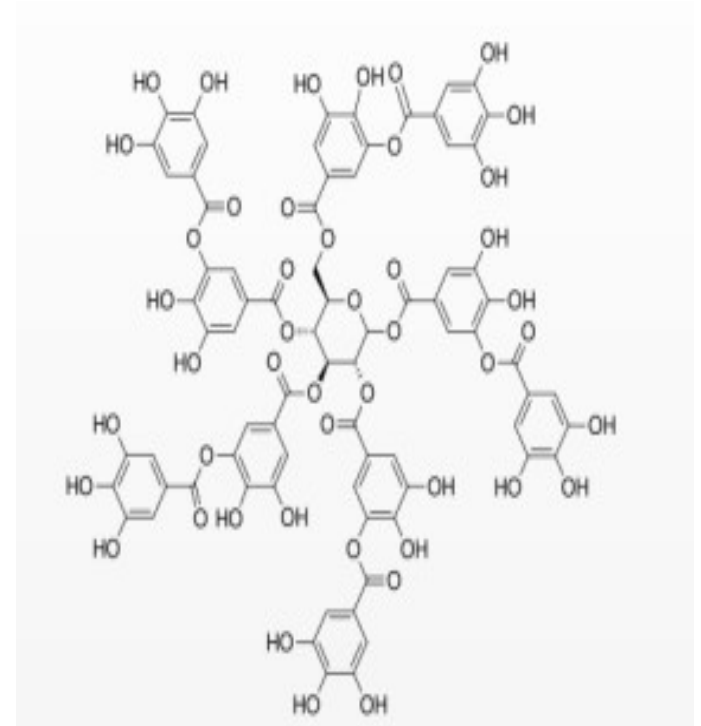
Hexahydroxydiphenic
acid



Ellagic acid

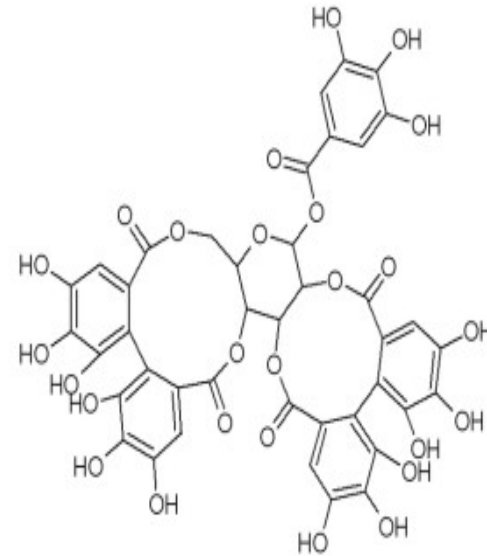
Gallitannins

- **galloyl groups** are linked by **depside bond**.
- A **depside** is a type of polyphenolic compound composed of two or more monocyclic aromatic units linked by an ester bond.
- Upon hydrolysis gallic acid is produced

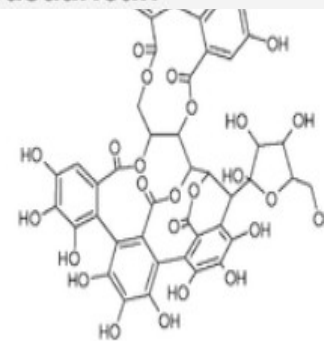


Elagitannins

- **galloyl groups** are linked through **C-C bonds**
- Upon hydrolysis ellagic acid is produced
- Slowly soluble in water
- Present in plants in open ring forms as hexa hydroxydiphenic acid



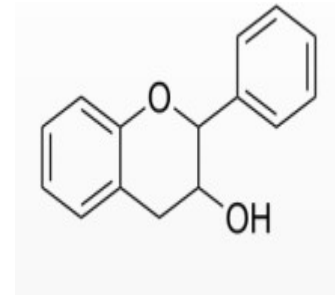
Casuarictin



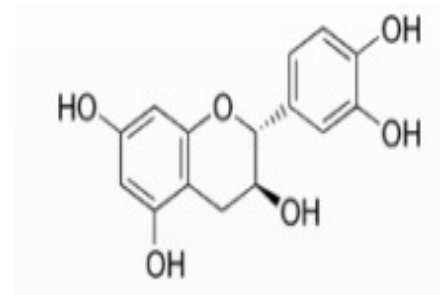
Grandinin

Condensed tannins (Catechol) (proanthocyanidins)

- Unlike hydrolysable tannins, these are **not readily hydrolyzable** to simpler molecules and
- they **do not contain a sugar moiety**.
- They are related to flavonoid pigments and have **polymeric flavan-3-ol structures** (usually di- and trimers).
- On **dry distillation they yield catechol** and these tannins are therefore sometimes called **catechol tannins**.

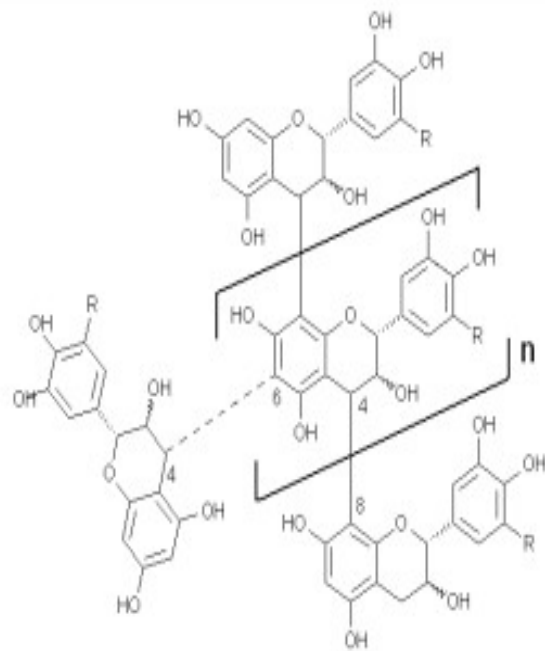


Flavan 3- ol



Catechin

- **Like catechol** itself, their solutions turn to **green with ferric chloride**.
- On **treatment with acids or enzymes condensed tannins** are converted into **red insoluble** compounds known as **phlobaphenes**.
- **Phlobaphenes** give the characteristic **red color** to many drugs such as red cinchona bark, which contain these phlobatannins and their decomposition products

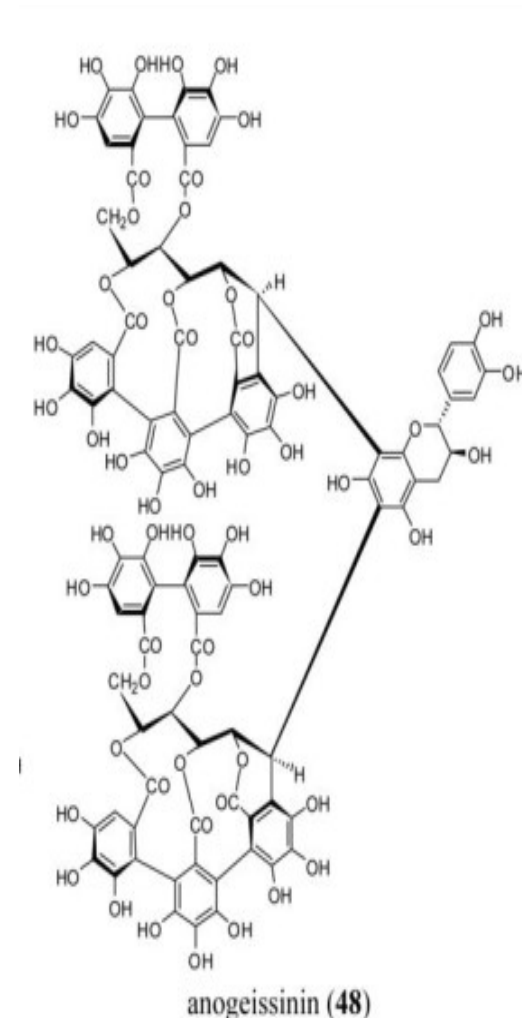


Condensed tannins can be linear with 4-8
 bounds OR
 Branched with 4-6 bounds

	Hydrolysable (Pyrogallol)	Condensed (Catechol)
Hydrolysis	Hydrolysable	Non-Hydrolysable
Dry distillation	Gives pyrogallol	Gives catechol
FeCl₃	Blue-black colour	Green colour
Br₂ water	No ppt.	Ppt
Examples	Gall, Clove, Rhubarb	Cinnamon, Tea, Henna.

Complex tannins

- These represent a group of **tannins that are biosynthesized from both hydrolysable tannin** (mostly a C-glucoside ellagitannin) (flavono-ellagitannin) and **condensed tannin**.
- The union occurs through a C—C bond between the C-1 of the glucose unit of the ellagitannin and the C-8 or C-6 of the flavan-3-ol derivative.



Pseudotannins

- are compounds of low molecular weight
- give some of the reactions of tannins
- usually found in combination with tannins

Examples:

- catechins
- chlorogenic acid
- ipecacuanic acid

Physical properties

- Tannins are **non-crystallizable** compounds
- Tannins are soluble in water, dilute alkalis, alcohol, glycerol and acetone, but generally only sparingly soluble in other organic solvents.
- Solutions precipitate heavy metals, alkaloids, glycosides and gelatin.
- With ferric salts, gallitannins and ellagitannins give **blue-black precipitates** and condensed tannins **brownish-green**.

Tests for identification

Goldbeater's skin test.

- Goldbeater's skin is a membrane prepared from the intestine of the ox and behaves similarly to an untanned hide
- Soak a small piece of goldbeater's skin in 2% hydrochloric acid;
- rinse with distilled water and place in the solution to be tested for 5 min.
- Wash with distilled water and transfer to a 1% solution of ferrous sulphate.
- A brown or black color on the skin denotes the presence of tannins.

Gelatin test:

- Solutions of true tannins (about 0.5-1%) precipitate a 1% solution of gelatin containing 10% sodium chloride.
- Gallic acid and other pseudotannins may react in concentrated solutions.

Quantitative methods

Hide powder method:

- This is the most widely used on commercial scale.
- A known amount of a reference sample of hide powder is added to the drug extract.
- The difference in the dry weight of the extract before and after treatment with hide powder is a measure for the tannin content.

Medicinal use

- as astringent (in superficial wounds they enhance tissue regeneration).
- as antidotes in poisoning of heavy metals, alkaloids and some glycosides
- for the protection of inflamed surfaces of mouth and throat
- Have gained attention in the last few years because of their demanding ability to treat a wide range of infections and abnormalities due to ability to denature microbial enzymes, adhesins, plasma membrane transport proteins, etc.
- Because of their property to bind with metals and proteins, tannins also prevent the growth of microbes through the deprivation of [metal ion](#) and substrate

- Tannins are used in **leather industry** to transform raw animal skin to leather due to their ability to cross-link with proteins.
- To be **effective for tannage**, the **polyphenol molecule** must be **neither too large** as it will be unable to enter the interstices between the collagen fibrils of the animal skin **nor too small** as it will be unable to cross-link between the protein molecules of adjacent fibrils at several points.